

Week 4 Report: Data Preprocessing

Project: Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

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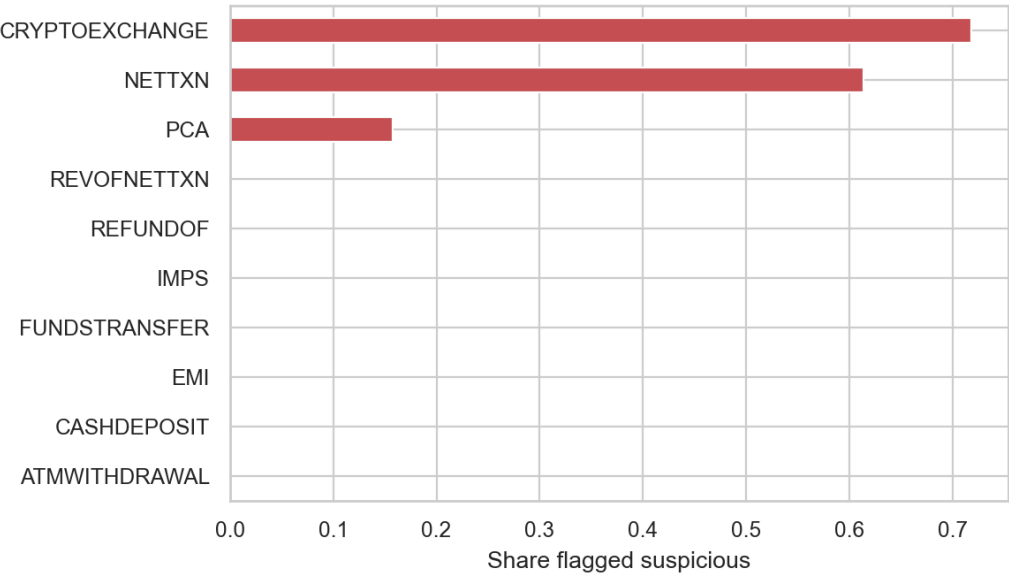
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What I did this week

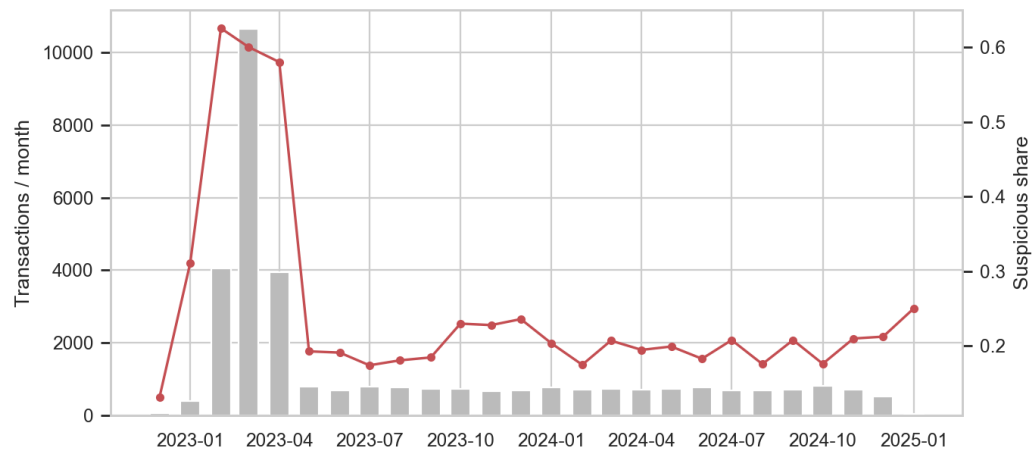
This week I explored the cleaned data. 14,203 of 33,611 transactions (42.3%) are flagged suspicious, and the data runs from 2022-12-16 to 2025-01-12.

The most useful finding was the channel. Seven channels, including ATM withdrawals, cash deposits, IMPS and fund transfers, are never flagged. Crypto exchanges are flagged in 72% of rows and NETTXN online payments in 61%. This means the labels are strongly tied to the channel, which I need to keep in mind when I judge how good the models really are.

Channel	Rows	Suspicious rate
NETTXN	16,070	61.4%
PCA	6,335	15.8%
CRYPTOEXCHANGE	4,653	71.8%
IMPS	3,003	0.0%
FUNDSTRANSFER	1,116	0.0%
REVOFNETTXN	506	0.0%
ATMWITHDRAWAL	500	0.0%
EMI	482	0.0%



Suspicious rate by channel.



Transactions and suspicious rate by month.

Next week

Train the first versions of Decision Tree, Random Forest and SVM, plus simple baselines.